GNAMMA RESEARCH · PRIMARY TEXTS

المنطق

al-Manṭiq

Muḥammad Riḍā al-Muẓaffar

Reading edition

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INTRODUCTION

Introduction to Logic

المدخل

INTRODUCTION

Why logic is needed

الحاجة إلى المنطق

God has made the human being naturally capable of speech and has given him a tongue as its instrument. Yet he still needs something to regulate and correct his speech, so that it accords with the language he learns in both the forms and materials of its expressions. He therefore needs, first, a teacher who habituates him to its use and, second, rules to consult that guard his tongue against error: these are grammar and morphology.

Likewise, God has endowed the human being with an intellective power and an aptitude for thought. Even so, people make many mistakes in thinking: they take something for a cause that is not a cause, regard as a consequence something that does not follow, and mistake what is not a demonstration for a demonstration. One may even hold a true conclusion on the strength of unsound premises. There is therefore a need for something that corrects thought, directs it towards valid inference, and trains it in the proper organisation of its operations.

Logicians describe logic as an instrument by which one can protect oneself against error and correct one's thinking. Grammar does not give one the ability to speak; it guides one to speak correctly. Similarly, logic does not impart the ability to think; it guides thought towards correctness.

Our need for logic, then, is the correction of our thinking—and how great that need is. If someone objects that people learn logic yet still reason badly, we answer that people learn grammar and morphology yet still make mistakes in speech. This happens because they have not acquired a settled mastery of the discipline, have failed to observe its rules when required, or have applied them incorrectly.

1. **Translator's note:** The distinction between the possession of a rational faculty and a trained habit (malaka) of correctly exercising it is central here. Logic is an instrumental discipline, not a substitute for intellectual attention.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Arabic text, Dār al-Taʿāruf (1402 AH) · p. 8–9 https://lib.rafed.net/view.php?b_id=1604&page=9&type=c_fbook

INTRODUCTION

Definition and instrumental character of logic

تعريف علم المنطق — المنطق آلة

Logic is accordingly defined as “a rule-governed instrument whose observance safeguards the mind against error in thinking.” Attend especially to the word *observance*. Learning logic does not automatically protect someone from mistakes, just as learning grammar does not automatically prevent errors of speech. The rules must be applied when they are needed.

Consider also the word *instrument*. Logic belongs to the instrumental sciences, which are employed for an end other than simply knowing their own propositions. It sets out the general, correct procedures by which thinking may attain truths not yet known, much as algebra investigates procedures for solving equations and arriving at unknown quantities.

More precisely, logic teaches general rules of correct reasoning applicable across the sciences. It investigates the structure and arrangement by which the mind moves from the intelligible contents already present to matters as yet unknown. Hence it has been called the *balance*, the *standard*, and the *servant of the sciences*. Even algebra, which served as our comparison, depends upon logical ordering in solving its problems.

A student of logic must therefore work exercises and actually employ its operations during study, as students of mathematics and the natural sciences must do.

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1. **Terminology:** Āla qānūniyya: an instrument governed by rules. The verb rāʿā (‘to observe, heed, apply’) is part of the definition; logic does not promise that mistakes become impossible merely by having studied a textbook.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Arabic text, Dār al-Taʿāruf (1402 AH) · p. 9–10 https://lib.rafed.net/view.php?b_id=1604&page=10&type=c_fbook

INTRODUCTION

From sensation to intellectual knowledge

العلم — تمهيد

We said that the human being is naturally disposed towards thinking and is capable of acquiring knowledge through an intellectual faculty that distinguishes him from non-rational animals. To clarify what we mean by knowledge and how logic is related to it, we may distinguish stages of cognitive development.

First, the newly born child's soul possesses an innate capacity but no acquired, actual knowledge. As sight, hearing, taste, smell, and touch begin to operate, the soul is affected by the things presented to these senses. This is sensory cognition, the first degree of knowledge and a beginning

from which further knowledge develops. Other animals share some or all of these sensory powers.

Second, the growing child begins to work upon retained sensory forms, comparing one with another—this object is taller, this light is brighter—and combining images into things not encountered externally. A town the child has only heard described may be imagined from previously seen towns. This is imaginative cognition (*ʿilm khayālī*), which some other animals also possess.

Third, cognition extends to particular meanings that are neither material nor measurable: the love shown by parents, the hostility of an enemy, the fear of a frightened person, the grief of someone bereaved, or the joy of a delighted person. This is estimative cognition (*ʿilm wahmī*). It is still not the distinctive perfection of the human intellect.

Fourth, the human being exercises intellect and reasoning over sensible, imaginative, and estimative contents. The intellect discriminates the true from the false, abstracts universal notions from particulars, compares these notions, and proceeds from what it knows to further judgments and conclusions. This is intellectual knowledge in its fullest sense.

The sciences and disciplines aim to cultivate this rational power. Logic, in particular, orders its activity and guards against its being misled by imagination or estimation.

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1. **Translator's note:** This passage is the author's graduated account of acquired cognition. The distinction between the faculty of estimation (wahm), imagination (khayāl), and intellect (ʿaql) is preserved rather than flattening all three into 'thinking'.
2. **Authorial footnote — source:** In the printed source Muẓaffar explicitly restricts the present discussion to ʿilm ḥuṣūlī (knowledge through an acquired form). Knowledge by presence (ʿilm ḥuḍūrī) differs because the known is present by its very existence, not through an impressed representation. Only the former is here divided into conception and assent.
3. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Arabic text, Dār al-Taʿāruf (1402 AH) · p. 10–12 https://lib.rafed.net/view.php?b_id=1604&page=11&type=c_fbook

INTRODUCTION

Knowledge as the presence of a form

تعريف العلم

One may ask how these acts of cognition arise in a human being. To make the preceding explanation clearer, look at something before you, then close your eyes and direct your attention towards it. You will find something in yourself as though you were still seeing it. Likewise, after hearing a clock ticking, block your ears but turn your attention to the sound; something of the hearing seems to remain. The experiment can be repeated with the other senses.

On examining these experiences, the author invites us to understand cognition as the reception of the forms of things in the soul, by analogy with the way an image is impressed in a mirror. On this account, knowledge is defined as “the presence of a thing's form to the intellect,” or, equivalently for present purposes, “its being impressed upon the intellect.”

1. **Interpretive note:** This is Muẓaffar's introductory presentation of representational or acquired knowledge, not a claim that every kind of knowledge is reducible to a material impression. See the preceding author's footnote on the distinct case of knowledge by presence.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Arabic text, Dār al-Taʿāruf (1402 AH) · p. 12–13 https://lib.rafed.net/view.php?b_id=1604&page=13&type=c_fbook

INTRODUCTION

Conception and assent

التصور والتصديق

Draw a triangle: its form comes to be present in the mind. This cognition is called *conception* (taṣawwur), considered here simply as an apprehension without affirmation or judgment. The same is true when you attend to its angles, or picture a horizontal line intersected by a perpendicular and thereby picture two right angles.

Now compare those two right angles with the sum of the triangle's angles. At first you may ask whether they are equal and remain uncertain. Even apprehending the *relation of equality* between them at this point is still mere conception.

Once you demonstrate their equality, however, a further cognitive state arises: you apprehend that the relation accords with reality, and this apprehension entails the soul's judgment and assent. This is called *assent* (taṣdīq), the cognition being named by its inseparable concomitant, the act of affirming that the relation obtains.

Thus the triangle's angles, the two right angles, and their proposed equality can all be conceived without judging. Assent occurs when the mind apprehends the relation as actually corresponding to reality. Recognising a proposition as *not* corresponding to reality is also a kind of assent—namely, a negative judgment.

Notice the qualification. In the broadest usage, conception, apprehension, and knowledge can all refer to the presence of a form to the intellect. Assent itself presupposes such an apprehension; but in the restricted contrast here, *conception* denotes apprehension without judgment, and *assent* denotes apprehension that entails judgment. Consequently, “conception” in the unrestricted sense may comprehend both.

1. **Authorial context:** Muẓaffar's accompanying note relates this account of taṣdīq to Avicenna's discussion in the Ishārāt and to Ṣadr al-Dīn al-Shīrāzī's independent treatise on conception and assent. He does not simply identify taṣdīq with an extra pictorial representation.

2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf Arabic witness · p. 13–15 https://lib.rafed.net/view.php?b_id=1604&page=14&type=c_fbook

INTRODUCTION

What can be conceived, and what can be assented to?

بماذا يتعلق التصديق والتصور؟

Assent has only one immediate object: the relation expressed in an assertoric sentence, insofar as the mind judges that relation to correspond—or not correspond—to reality.

Conception, by contrast, can concern four things. First, a single expression: a noun, verb, or particle. Second, the relation asserted in a statement when we are uncertain of it or merely entertain it, as when we consider the possibility that Mars is inhabited without affirming it. Third, the relation expressed by an injunction, prohibition, wish, question, or other non-assertoric utterance; as such, these are not descriptions to be affirmed or denied as corresponding to reality. Fourth, an incomplete expression, such as a genitive phrase, a noun and its adjective, a relative clause, or one member of a conditional construction.

Take the Qurʾānic conditional, “If you count God's blessings, you cannot enumerate them.” Each of its components is apprehended as a component of the conditional, rather than as a separately asserted proposition. Likewise, “God's blessings” is an incomplete genitive expression. The conditional statement taken as a whole, however, is something to which assent may be given.

The logical distinction here concerns the *role in which* an expression is apprehended, not merely whether it contains words capable of forming a statement outside that construction.

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1. **Translator's note:** Muẓaffar uses ‘maʿlūm taṣawwurī’ for a known content apprehended without assent. The conditional's antecedent and consequent do not become independently asserted merely by occurring within a larger proposition.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf Arabic witness · p. 15–16 https://lib.rafed.net/view.php?b_id=1604&page=16&type=c_fbook

INTRODUCTION

Certainty, opinion, doubt, and counter-opinion

أقسام التصديق

Assent is divided into *certainty* (yaqīn) and *probable opinion* (ẓann). It consists in preferring one of the two sides of a report—its obtaining or not obtaining. If the mind excludes the alternative altogether, this is certainty; if the alternative remains possible, though less likely, it is opinion.

Consider any report. Either you admit only one possibility, or you regard both alternatives as possible. The first condition is certainty. The second admits three possibilities: the alternatives may be equally balanced, in which case the state is *doubt* (shakk); the content of the report may be preferred, yielding probable opinion (ẓann); or the opposite alternative may be preferred, in which case the report's content is only a weaker apprehension (wahm).

There are thus four states, not five. (1) **Certainty:** you affirm the report and rule out its falsity, or deny it and rule out its truth. (2) **Opinion:** you favor one side while admitting the other as possible. (3) **Counter-opinion:** you entertain one side while favoring its opposite. (4) **Doubt:** you judge the alternatives equally possible.

Two consequences follow. In the classificatory sense used here, doubt and the weaker opinion are not themselves assent; and probable opinion about one side is always the weaker opinion concerning the opposing side. The corresponding terms are relational to whichever side of the proposition is being considered.

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1. **Terminology:** Wahm here denotes the weaker side of a propositional assessment; it is not simply the psychological faculty of estimation described earlier. The identical word is used in two technical contexts.
2. **Authorial note:** The printed apparatus distinguishes a narrower sense of yaqīn: firm, factually correct assent that is not merely accepted by imitation. The chapter's initial division uses the broader sense of firm conviction.
3. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf Arabic witness · p. 16–17 https://lib.rafed.net/view.php?b_id=1604&page=17&type=c_fbook

INTRODUCTION

Simple and compound ignorance

الجهل وأقسامه

Ignorance is the absence of knowledge in a subject capable of knowing. We do not call a stone blind, because blindness is the lack of sight in something naturally fitted to see. Similarly, knowledge and ignorance are opposites in the manner of a possession and its privation.

Since knowledge is distinguished into conception and assent, one may speak correspondingly of ignorance affecting conception and ignorance affecting assent. The further distinction between *simple* and *compound* ignorance, however, strictly concerns the latter: ignorance in conception can only be simple.

Simple ignorance is not knowing something while recognising that one does not know it. Compound ignorance occurs when a person does not know something yet believes that he does know it. It comprises two deficiencies: ignorance of the thing and ignorance of that ignorance. Because it includes a mistaken conviction, it belongs to the sphere of assent.

Compound ignorance, Muẓaffar argues, should not be counted as genuine knowledge merely because it includes a firm belief. On his definition, knowledge presents the form of the very thing known. Mistaking a different form for it does not make that form the thing itself. A distant silhouette taken for a human being does not become a human being through the observer's conviction.

The point is about the truth-condition implicit in this account of knowledge, not an assertion that everyone who holds a mistaken belief knows nothing whatever.

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1. **Context:** This passage is useful for distinguishing doxastic firmness from knowledge: a firmly held false conviction remains compound ignorance rather than certainty in the narrower epistemic sense.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf Arabic witness · p. 17–18 https://lib.rafed.net/view.php?b_id=1604&page=18&type=c_fbook

INTRODUCTION

Immediate and acquired knowledge

العلم ضروري ونظري — توضيح في الضروري

Both conception and assent are divided into two kinds. The first is *immediate* or self-evident (*ḍarūrī, badīhī*): it does not require a further process of inquiry or discursive thought. Examples include apprehending the concepts of being and non-being and assenting that the whole is greater than its part or that contradictories cannot both obtain.

The second is *discursive* or acquired (*naẓarī, kasbī*): it requires a process of thought. A conclusion about the movement of the earth, or a conceptual account of an unfamiliar phenomenon, may need prior knowledge to be arranged and used in an inquiry.

For immediate cognition it is sufficient that the mind attend to the relevant object without an intervening inferential process. For discursive cognition the mind must perform intellectual operations, much as one manipulates the terms of an algebraic problem, moving from what is already known to the initially unknown.

Self-evident, however, does not mean that every person actually knows every self-evident thing. The disposition of the subject and the conditions of attention matter. Muẓaffar distinguishes five conditions or occasions for the soul's attending to such knowledge: (1) attention rather than distraction, (2) a sound cognitive faculty, (3) functional senses when sense perception is involved, (4) freedom from a misleading sophism, and (5) some non-inferential operation appropriate to the kind of knowledge, such as receiving convergent testimony or undertaking an experiment.

The classification therefore concerns whether the knowledge depends on an *inferential act of thought*, not simply whether attaining it took time or effort.

1. **Translator's note:** Do not translate ḍarūrī in this context as ‘necessary’ in the metaphysical modal sense. The distinction is between epistemic immediacy and discursive acquisition.
2. **Translator's note:** Muẓaffar's fourth condition is especially significant for kalām: a specious argument may obscure an otherwise evident principle, such as the impossibility of simultaneous affirmation and denial. He describes this as a sophism in opposition to an evident truth.
3. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf Arabic witness · p. 18–20 https://lib.rafed.net/view.php?b_id=1604&page=19&type=c_fbook

INTRODUCTION

Thought as the intellect's movement between known and unknown

تعريف النظر أو الفكر

By inquiry (*naẓar*) or thinking (*fikr*), Muẓaffar means the performance of an intellectual operation upon what is already present to the mind for the sake of attaining the object sought—knowledge of something previously unknown. More precisely, thought is “the movement of the intellect between the known and the unknown.”

When faced with a problem, one first identifies what kind of unknown it is, then turns to the stock of knowledge relevant to that kind of problem. The intellect searches and compares these contents, arranging them until it finds a combination capable of resolving the difficulty. It then moves from this ordered knowledge to the desired conclusion.

The process may be analysed into five moments: (1) encountering the problem; (2) identifying its type; (3) moving from the problem to stored knowledge; (4) examining and combining that knowledge appropriately; and (5) moving from the assembled premises to the answer. The last three movements, strictly speaking, constitute discursive thought.

These stages often occur so swiftly that we are not consciously aware of them. In someone gifted with strong *ḥads* (intuitive insight), the first two discursive movements can be bypassed: the relevant knowledge yields the conclusion in a single immediate movement. This helps explain why a proposition may be self-evident for one mind but require discursive inquiry for another.

Here ‘movement’ names the ordered activity of cognition; it should not be confused with physical locomotion.

1. **Terminology:** Ḥads in this passage signifies an immediate intellectual grasp that dispenses with some intermediate search. It should not be assimilated without qualification to the separate notion of mystical witnessing.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf Arabic witness · p. 20–22 https://lib.rafed.net/view.php?b_id=1604&page=21&type=c_fbook

INTRODUCTION

The six principal divisions of logic

Logic is needed above all to acquire discursive knowledge, because it provides the rules governing inquiry and thought. Immediate truths are available without discursive proof and serve as the initial capital from which further knowledge is obtained. As knowledge increases, the stock of what is already known grows, and further investigation becomes possible.

Logic therefore studies how to combine available cognitions so as to gain cognitions not yet possessed. When the unknown is conceptual, the route is through a *defining account* (muʿarrif); when the unknown is a judgment, the route is through an *argument* (ḥujja). The investigation of arguments may concern either the form of their composition or the character of their premises—the latter treated under the five argumentative arts.

The book is accordingly arranged into six parts: (1) expressions and signification; (2) universals; (3) definition and division; (4) propositions and their properties; (5) arguments and their formal composition; and (6) the five arts.

The architecture matters: the first three parts concern how to attain a previously unknown conception; the later parts explain how to attain or evaluate previously unknown judgments.

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1. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf Arabic witness · p. 23–25 https://lib.rafed.net/view.php?b_id=1604&page=24&type=c_fbook

STAGE L

Book I · Expressions and Signification

الباب الأول — مباحث الألفاظ

CHAPTER 1

Why logic must investigate expressions

الحاجة إلى مباحث الألفاظ

The logician's principal concern is with intelligible meanings themselves. Nevertheless, the examination of expressions cannot be dispensed with, because expressions provide a means of reaching those meanings. People ordinarily communicate and exchange thoughts through one language or another, and words can undergo alteration, ambiguity, or confusion in ways that obstruct understanding. Logic therefore considers the conditions of expressions in general, rather than the peculiarities of one language, so that we can assess our own speech and that of others by an appropriate standard.

The qualification *in general* is important: logic is not a discipline restricted to speakers of a particular language. Sometimes a logical inquiry may require a question belonging to the grammar of the language in which it is expressed—for example, whether the Arabic definite article has a universal or exhaustive import, the logical role of *kāna* and related expressions, or the particles expressing generality and negation. But such matters can often be left to the sciences of language rather than incorporated into logic itself.

Thus inquiry into words is instrumental: it safeguards the passage between our linguistic expressions and the meanings that are the proper objects of logical investigation.

1. **Translator's note:** The distinction is between the logician's primary object (intelligible meanings) and the necessary instrumental investigation of language. Rules peculiar to Arabic grammar are not automatically universal laws of logic.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Arabic text · p. 29–32 https://lib.rafed.net/m/?b_id=1604&type=c_fbook

CHAPTER 2

Signification: rational, natural and conventional

تعريف الدلالة وأقسامها

When you hear a knock at your door, your mind passes to the thought that someone outside is calling for you. The sound of the knock has disclosed the presence of that person. We say that it *signifies* the person's presence.

The knocking is the **sign** (*dāll*), the presence of the person is the **signified** (*madlūl*), and the sign's having this property is **signification** (*dalāla*). In general, whenever knowing one thing leads the mind to the existence of another, the first is a sign and the second what is signified. Signification is thus the condition of a thing whereby awareness of it leads the mind to another thing.

This intellectual transition cannot occur without an explanation. It depends upon a firmly established connection between the two items in the mind, itself grounded in awareness of a connection or concomitance between them. Because the connection can be essential, natural, or instituted by convention, signification is divided into three kinds.

1. Rational signification (dalāla ʿaqliyya). The sign and the signified have an intrinsic connection in their external existence, as effect to cause. When one knows that daylight follows the rising sun and sees the morning light on a wall, one infers that the sun has risen. Similarly, hearing a voice behind a wall signifies the presence of someone speaking. The inference rests upon a known causal or existential connection.

2. Natural signification (dalāla ṭabʿiyya). The connection arises from a disposition or customary reaction of the human constitution and need not occur invariably in every person. Some utter ‘ouch’ when in pain, ‘ah’ when distressed, or a sound of disapproval when annoyed; others crack their knuckles, stretch, fidget with an object or their beard, make certain gestures while thinking, or yawn when drowsy. One who has learned such patterns may infer a person's pain or mental preoccupation from an utterance or gesture. Unlike the causal concomitance invoked in rational signification, these dispositions can vary across individuals.

3. Conventional signification (dalāla waḍʿiyya). Here the connection is established through an agreement or convention that one thing is to serve as a sign of another. Examples include writing as a sign of spoken words, gestures used instead of speech, communication signals, mathematical and geometrical symbols, symbols in other sciences, and words instituted to express intentions. When the convention is known and the sign is apprehended, the mind passes to what it signifies.

The division therefore distinguishes **the ground of signification**, not merely the physical kind of sign. A sound or gesture may function differently according to whether the relevant link is causal, natural, or conventional.

1. **Translator's note:** A ‘natural’ sign (ṭabʿī) is a disposition-dependent cue, potentially varying by person; a ‘rational’ sign (ʿaqlī) is inferred from an intrinsic causal or concomitant relation. This should not be conflated with the modern linguistic contrast between arbitrary and iconic signs.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Arabic text · p. 32–34 https://lib.rafed.net/m/?b_id=1604&type=c_fbook

CHAPTER 3

Verbal signification: total, partial, and concomitant

أقسام الدلالة اللفظية: المطابقية والتضمنية والالتزامية

Conventional signification is divided into **verbal** signification, where the instituted sign is an expression or word, and **non-verbal** signification, where the instituted sign is something other than a word—gestures, written marks, scientific symbols, road signs indicating distance or direction, and similar conventions.

The reason a word signifies a meaning is the established intellectual connection between expression and meaning. That connection arises from convention, insofar as the speakers know the convention. Verbal signification is therefore the condition of an expression through which knowing that the speaker has uttered it gives us knowledge of the meaning intended by the speaker.

A word signifies in three different respects.

1. Total or complete signification (dalāla muṭābaqiyya). The word signifies the whole of the meaning for which it was instituted. ‘Book’, for instance, signifies the complete object including its pages, markings and cover. ‘Human’ signifies the whole defining account ‘rational animal’. This is the basic signification for which the expression was instituted.

2. Inclusive or partial signification (dalāla taḍammuniyya). The word signifies an essential part included within its full instituted meaning. ‘Book’ signifies its pages or its cover; ‘human’ signifies animality or rationality as a constituent of its definition. A purchaser of a book understands its cover to be included. If the seller later attempted to exclude it, the purchaser could appeal to what was contained in the meaning of ‘book’. Partial signification depends upon total signification because the part is understood through the prior meaning of the whole.

3. Concomitant or implicative signification (dalāla iltizāmiyya). The expression leads the mind to a meaning **outside** its instituted meaning that is nevertheless connected to it as a necessary or familiar concomitant. ‘Inkwell’ may imply a pen. If someone requested an inkwell and the respondent brought no pen, the requester might appeal to the understood implication. This implicative signification also depends upon total signification: to infer something associated with a meaning, the mind must first understand the meaning actually signified by the expression.

The distinctions concern whether the signified content is the **whole**, an **internal constituent**, or an **external concomitant** of the word's instituted meaning. They should not be confused with the speaker's intention to assert some particular content.

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1. **Authorial apparatus · paraphrase:** The author explains why the verbal/non-verbal division is emphasised for conventional signification: although rational and natural signs may themselves involve sounds or words, the distinction has special logical utility for conventional verbal signification and its three forms.
2. **Terminological note:** Taḍammun is the signification of a part included within the whole instituted meaning; iltizām is signification of what is outside the instituted meaning yet connected to it. Do not render both simply as implication.
3. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Arabic text · p. 34–36 https://lib.rafed.net/m/?b_id=1604&type=c_fbook

CHAPTER 4

The condition for implicative signification

شرط الدلالة الالتزامية

Implicative signification requires a **mental** connection between the instituted meaning of the expression and the further meaning that follows from it. A concomitance existing only externally is insufficient if it has not been established in the mind: without that, the mind would not pass from the expression's meaning to the further meaning.

The connection must also be evident or immediately apparent. On apprehending the meaning of the expression, the intellect must be able to pass to its concomitant **without needing a further mediating step**.

Thus a logical or physical association, however strong, does not by itself constitute signification by implication for someone who has never grasped the relation. The necessary condition is an intelligible and evident mental concomitance.

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1. **Translator's note:** The particle ‘also’ (أيضاً) adds a separate condition: not only must the relation be mental, it must be sufficiently evident to produce the transition without further mediation. In usūl al-fiqh and later logical discussions the precise scope of ‘evident concomitance’ may be further distinguished.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Arabic text · p. 36 https://lib.rafed.net/m/?b_id=1604&type=c_fbook

STAGE I

The Four Questions

مطلب ما وأي وهل ولم

CHAPTER 1

The first question: explaining a word

المرحلة الأولى: التعريف اللفظي

When an expression from any language confronts you, there are five successive stages through which you must pass to attain knowledge. Some seek knowledge by way of conception, and others knowledge by way of assent.

The first stage is to seek an initial, general conception of the expression's meaning. If you do not know which meaning the word has been assigned, you ask a purely linguistic question. The answer is another word signifying that meaning. Thus, if you ask what ghaḍanfar means, the answer is lion; or what sumaydaʿ means, the answer is chief. An answer of this kind is called a verbal definition (taʿrīf lafẓī). Providing such definitions is the responsibility of dictionaries.

1. **Translator's note:** A verbal definition establishes customary signification. It need not yet explain the thing's essential constituents.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Book III (al-Muʿarrif) · p. 110 https://lib.rafed.net/view.php?b_id=1604&type=c_fbook

CHAPTER 2

What is it? The explanatory question

المرحلة الثانية: مطلب ما الشارحة

Once you have conceived the word's meaning in a general way, you turn naturally to the second stage: seeking a conception of the meaning's quiddity. That is, you seek an articulation of what the name had indicated only generally, so that the object may be distinguished completely from other objects in the mind. You therefore ask “What is it?” (mā huwa?).

The interrogative mā here is called explanatory (shāriḥa), because it asks for an explanation of what a word signifies. Its answer is called an explanation of the name (sharḥ al-ism), or nominal definition. The primary form of answer combines the proximate genus with the proximate differentia; this is the complete nominal essential definition. One may also answer with the

differentia alone, the proprium alone, either combined with a more remote genus, or the proprium together with the proximate genus. According to the constituents supplied, such accounts are called incomplete essential definitions or complete/incomplete descriptions; but at this stage all remain nominal.

Suppose, however, that a respondent gives only the proximate genus. Asked “What is a date-palm?”, he answers “A tree.” This will not satisfy the questioner, who still wants to know what distinguishes this tree from others. He asks “Which kind of tree is it, essentially?” or “Which kind is it by its distinctive attribute?” The former seeks the differentia; the latter seeks a proprium. This is the proper place of the interrogative “which?” (ayy).

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1. **Translator's note:** Muẓaffar's ‘nominal’ does not mean an arbitrary stipulation: the same account can become a real definition once existence has been established.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Book III (al-Muʿarrif) · p. 110–111 https://lib.rafed.net/view.php?b_id=1604&type=c_fbook

CHAPTER 3

Does it exist? Simple and composite hal

المرحلتان الثالثة والرابعة: هل البسيطة وهل المركبة

After learning what the expression means, you move to the third stage: seeking assent to the thing's existence. You ask with hal, the simple existential question (hal basīṭa): “Does such-and-such exist?” or “Is it existent?”

A qualification is necessary. The second and third stages can occur in either order. Their natural order is the one just given, but the questioner may already know that a thing exists, or may ask about its existence first. If existence is known before the general expression has been explained in detail, a subsequent question “What is it?” is called the real or quiddative ‘what?’ (mā ḥaqīqiyya). Its answer is the same as to the explanatory mā. The only difference is that the explanatory question precedes knowledge of the thing's existence, while the real question follows it. Hence an essential definition can be nominal before the simple existential judgment and real afterwards.

Once these stages are completed, one naturally passes to the fourth stage, seeking assent that some attribute or condition belongs to the thing. This is again asked by hal, but now it is the composite question (hal murakkaba), because it asks whether one thing is established of another, presupposing the subject's existence. By contrast, simple hal asks only whether the subject exists. “Does God exist?” is a simple existential inquiry; “Is the existent God willing?” is composite.

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1. **Translator's note:** ‘Simple’ and ‘composite’ distinguish asserting that a subject exists from predicating something further of an existent subject. Neither question by itself provides an essential definition.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Book III (al-Muʿarrif) · p. 111–113 https://lib.rafed.net/view.php?b_id=1604&type=c_fbook

CHAPTER 4

Why? The cause of assent and the cause of being

المرحلة الخامسة: مطلب لم

If a respondent has answered either the simple or composite hal, you proceed to the fifth stage: seeking the cause. Sometimes you seek only the ground of the judgment—namely, a demonstration of what the respondent affirmed. At other times you seek both the ground of the judgment and the cause of the fact's existence, so that you know why the thing is actually so.

For either purpose you ask “Why?” (limā?). To seek the ground of an affirmation, for instance, you may ask, “Why is God willing?” To seek both the reason for the judgment and the real cause, you may ask, “Why does a magnet attract iron?” Once someone has answered “Yes” to whether a magnet attracts iron, you are entitled to ask the further question, “Why?”

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1. **Translator's note:** The two senses anticipate the difference between demonstrating that a proposition is true and demonstrating why the fact it reports obtains.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Book III (al-Muʿarrif) · p. 112–113 https://lib.rafed.net/view.php?b_id=1604&type=c_fbook

CHAPTER 5

Summary of the inquiries and their subsidiary forms

تلخيص وتعقيب وفروع المطالب

We may summarise the foregoing as follows. “What?” (mā) seeks a conception of the thing's quiddity; it may be explanatory or real. From this interrogative the technical expressions māʾiyya and māhiyya are formed.

“Which?” (ayy) asks how a thing is distinguished, essentially or accidentally, from others sharing its genus, once that genus is known.

“Whether?” (hal) can be simple, seeking assent to the existence or nonexistence of the thing, or composite, seeking assent that something belongs—or does not belong—to something else.

“Why?” (limā) asks either for the cause of the assent alone, or for both the cause of the assent and the cause of the thing's existence.

These are the principal inquiries, whose questions have a universal scope across the sciences. Other inquiries employ “how?”, “where?”, “when?”, “how many?”, and “who?”. These are subsidiary, because they are not applicable to every object: what has no spatial location cannot be asked “where?”, and what has no time cannot be asked “when?”. Moreover, many such questions can be recast as composite hal questions.

1. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Book III (al-Muʿarrif) · p. 113–114 https://lib.rafed.net/view.php?b_id=1604&type=c_fbook

STAGE II

The Defining Account

المعرّف وأقسامه

CHAPTER 1

What is a defining account?

التعريف وأقسامه

Disputes in scientific and other matters frequently arise because the concepts conveyed by the words used remain indeterminate. Mutual understanding is disturbed when the disputants have not agreed on the boundaries of the words' meanings.

Anyone pursuing realities must therefore express the elements of his thought in language that defines and explains them clearly, preserving the intended meaning without exceeding or falling short of it. Sound thinking is built on this basis.

A defining account may be an essential definition (ḥadd) or a description (rasm); either may be complete or incomplete. We have already mentioned verbal definition. Here we are not primarily concerned with it, since it tells us only the meaning assigned to a word. The logician's subject is the previously known conception by which we arrive at a conception not yet known—the kind of account given in answer to explanatory or real mā.

1. **Translator's note:** The Arabic contrast is between ḥadd (essential definition), rasm (description through distinguishing attributes), and taʿrīf lafẓī (lexical explanation).
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Book III (al-Muʿarrif) · p. 115–116 https://lib.rafed.net/view.php?b_id=1604&type=c_fbook

CHAPTER 2

The complete essential definition

الحد التام

A complete essential definition states all the essential constituents of what is being defined. It is given through the proximate genus and proximate differentia, because together they encompass all those constituents.

If one asks, “What is a human being?”, the answer may be “A rational animal.” This is complete: it articulates what the word “human” signified without detail and includes all the essential constituents. For “animal” already implies substance, growing body, sensation, and movement by will.

One might instead answer, “A growing body that senses, moves voluntarily, and reasons.” This is also a complete definition, identical in meaning to the first but more fully expanded: the defining account of “animal” has been substituted for the word itself. Such expansion is unnecessary unless the questioner does not understand the quiddity of an animal.

Likewise, an unfamiliar genus may be unfolded by its own definition, if one is available, until the explanation reaches primary and self-evident concepts that require no definition—for example being and thing.

Hence the proximate genus and proximate differentia encompass every essential constituent without omission. A longer complete definition differs from a shorter one in explicitness, not in its intelligible content. The complete definition is conceptually equivalent to what it defines and signifies it directly, by correspondence (muṭābaqa).

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1. **Translator's note:** ‘Being’ (mawjūd) is named here as an immediately understood concept, not as a constituent that must be added to the essential definition of each contingent quiddity.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Book III (al-Muʿarrif) · p. 117–118 https://lib.rafed.net/view.php?b_id=1604&type=c_fbook

CHAPTER 3

The incomplete essential definition

الحد الناقص

An incomplete essential definition gives only some of the essential constituents of the thing, but must contain at least its proximate differentia. It therefore may be expressed through a remote genus together with the proximate differentia, or through that differentia alone.

In the first case, one might define a human being as “a growing body … rational,” omitting the intervening determination “sentient and moving by will.” In the second, one might say simply “rational,” dropping the entire proximate genus. The second is even less complete.

Three points follow. First, the incomplete definition is not conceptually equivalent to what it defines, because it states only part of its essential content, although its extension may coincide with the thing being defined. Second, it does not supply a complete mental conception of the

thing's quiddity. At most it distinguishes the thing essentially from everything else. Third, it signifies the whole by necessary implication (iltizām) rather than by direct correspondence, since a distinctive essential part points to the whole.

1. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Book III (al-Muʿarrif) · p. 118 https://lib.rafed.net/view.php?b_id=1604&type=c_fbook

CHAPTER 4

Complete and incomplete descriptions

الرسم التام والرسم الناقص

A complete description (rasm tāmm) combines the genus with a proprium (khāṣṣa): for example, “a laughing animal” as an account of the human being. It includes an essential constituent—the genus—and a distinctive but nonessential attribute—the proprium. For this reason it is called complete.

An incomplete description (rasm nāqiṣ) employs the proprium alone, as when a human being is described as “laughing” or “capable of laughter.” Because it contains only an accidental predicate, it is incomplete.

According to another formulation, genus plus proprium counts as complete only when the genus is the proximate one; a remote genus plus proprium remains an incomplete description.

In either case, a description, like an incomplete essential definition, serves to distinguish its subject from all other things. But it does so through what is extrinsic to the quiddity. It agrees with what it describes in the things to which it applies, not in the meaning it articulates, and points to it by implication rather than by a statement of the complete essence.

1. **Translator's note:** Here ‘accidental’ (ʿaraḍī) means extrinsic to the quiddity. A proprium need not be temporally separable or contingent in the ordinary English sense.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Book III (al-Muʿarrif) · p. 119 https://lib.rafed.net/view.php?b_id=1604&type=c_fbook

CHAPTER 5

Why definitions are difficult

إنارة

The primary form of definition is the complete essential definition, for the aim of defining is twofold: first, to conceive the thing through its true quiddity, so that a clear and fully articulated representation of it arises in the soul; second, to distinguish it completely from everything else in the mind. Only a complete essential definition achieves both aims.

Where the first aim is unattainable, one must be content with the second. An incomplete essential definition or either kind of description may provide it. If an essential distinction is available, an incomplete essential definition is preferable to a description; among descriptions, the complete is preferable to the incomplete.

Yet scholars acknowledge that grasping things' true essences and differentiæ is extraordinarily difficult and may be unattainable. Many of the differentiæ customarily given are really necessary proprieties that disclose the true differentia without themselves being that differentia. Thus most—or perhaps all—of the definitions available to us are descriptions resembling essential definitions.

Accordingly, one should select an evident and necessary distinctive attribute, one that best reveals the thing's reality and most closely resembles its differentia. An obscure attribute will not serve to explain the thing to an ordinary questioner.

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1. **Translator's note:** The passage itself qualifies the theory of ḥadd: satisfying the formal schema ‘genus + differentia’ does not mean the alleged differentia has actually been shown to be the real one.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Book III (al-Muʿarrif) · p. 120 https://lib.rafed.net/view.php?b_id=1604&type=c_fbook

CHAPTER 6

Definitions by example and by comparison

التعريف بالمثال والتشبيه

Scholars, especially those concerned with language and literature, often explain a thing by citing one of its instances as an example. This is definition by example, and it is particularly accessible to beginners seeking to understand and distinguish things.

A related educational procedure is inductive instruction: before stating the definition or general rule, the author or teacher supplies many examples and exercises, allowing the student to infer the common conception or rule independently. Only then is the result expressed clearly, so that the learner can compare the explicit statement with what he has discovered.

Definition by example is not a fifth fundamental kind of definition. It is subordinate to explaining something through a distinctive feature, and hence falls under incomplete description.

Similarly, definition by resemblance explains an unfamiliar thing by comparing it with something already known through a relevant likeness. An example is the comparison of existence with light: each is manifest in itself and makes other things manifest.

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1. **Translator's note:** The light comparison illustrates a point; it is not itself a proof that existence is literally a species of light.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Book III (al-Muʿarrif) · p. 121–122 https://lib.rafed.net/view.php?b_id=1604&type=c_fbook

STAGE III

Conditions of Definition

شروط التعريف

CHAPTER 1

A definition must be coextensive and clearer

شروط التعريف: الأول والثاني

The point of a definition is to make its subject intelligible and distinguish it from everything else. Five conditions are therefore required.

First, the defining account must be coextensive with what it defines. It must be both exclusive (māniʿ) and inclusive (jāmiʿ): exclusive because it applies to nothing outside the subject's extension, and inclusive because it applies to every instance of the subject without exception.

Accordingly, an account too broad is invalid: “a two-legged animal” does not exclusively identify the human being, since other animals walk on two legs. An account too narrow is likewise invalid: “an educated animal” excludes many human beings. Nor may a completely disparate description define the subject, because the two do not apply to the same instances.

Second, the defining expression must be clearer and better known to the addressee than what is being defined. Otherwise it fails to explain. One should not explain an unknown term through something equally obscure—for example by defining one of two correlatives solely through the other when both are equally unfamiliar—nor through something still more obscure, as though explaining light by saying that it is a power resembling existence.

1. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Book III (al-Muʿarrif) · p. 123–124 https://lib.rafed.net/view.php?b_id=1604&type=c_fbook

CHAPTER 2

No self-definition and no circularity

شروط التعريف: الثالث والرابع

Third, the defining account must not be identical in meaning with its subject, when a real rather than a merely verbal definition is intended. To explain “motion” by the synonymous “transition,” or “human being” by the synonymous “person,” provides no new intelligible determination. The two expressions must differ either as an implicit content differs from its explicit analysis, as in a complete essential definition, or by a distinct conception which makes the subject known. Otherwise knowledge of the subject would be presupposed in the very attempt to acquire it.

Fourth, the account must not be circular. Circularity arises when the expression offered to explain the subject is itself unintelligible except through that subject. The subject then becomes part of its own explanation.

Explicit circularity occurs when the sun is defined as a star that rises during the day, while day is defined as the time during which the sun rises. Knowing the sun presupposes knowing day, which in turn presupposes knowing the sun.

Concealed circularity may involve several mediating steps. Suppose “two” is explained as the first even number, “even” through division into equals, “equals” through two matching things, and “two things” through the number two. The circuit eventually returns to the very notion it was meant to explain.

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1. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Book III (al-Muʿarrif) · p. 124–126 https://lib.rafed.net/view.php?b_id=1604&type=c_fbook

CHAPTER 3

Unambiguous language in definitions

شروط التعريف: الخامس

Fifth, a definition must be expressed in clear, unambiguous language. One should not employ obscure, rare, or unfamiliar expressions, nor equivocal or figurative terms without an indication fixing the intended sense. Where such an indication is present, the use of these terms is not formally prohibited; nevertheless, scientific definitions and argument should generally avoid metaphor when greater precision is available.

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1. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Book III (al-Muʿarrif) · p. 126 https://lib.rafed.net/view.php?b_id=1604&type=c_fbook

29 original passages; 29 with working English. Not a specialist-certified critical edition.